Monte Carlo simulation of Heston model

Gamma Simulation and Three Numerical Schemes

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Heston Model for Monte Carlo Pricing

Model setup

- At-the-money European call: $K = S_0 = 1$, maturity $T = 1$.
- Initial variance: $v_0 = 0.1$; risk-free rate: $r = 0$.
- Goal: compare three Monte Carlo schemes under stochastic volatility.

Heston dynamics

$$
\begin{aligned}
dS_t &= rS_t\,dt + \sqrt{v_t}\,S_t\,d\hat{Z}_t,\\
dv_t &= \kappa(\theta - v_t)\,dt + \sigma\sqrt{v_t}\,dW_t,\\
d\hat{Z}_t &= \rho\,dW_t + \sqrt{1-\rho^2}\,dZ_t,
\end{aligned}
$$

where W_t and Z_t are independent Brownian motions.

Numerical focus

- Preserve non-negativity and stability of v_t.
- Control time-discretization bias in S_t while keeping GPU efficiency.

Exact CIR Variance Transition and Gamma Sampling

Why Gamma sampling?

- The CIR process admits an *exact* transition law (non-central χ^2).
- This is decomposed into a Poisson–Gamma mixture (cf. [4], [8]).

Exact transition (Step 1)

$$d = \frac{2\kappa\theta}{\sigma^2}, \quad \lambda = \frac{2\kappa\, e^{-\kappa\Delta t}\, v_t}{\sigma^2(1 - e^{-\kappa\Delta t})}, \quad N \sim \mathcal{P}(\lambda), \quad v_{t+\Delta t} = \frac{\sigma^2(1 - e^{-\kappa\Delta t})}{2\kappa}\, \mathcal{G}(d + N).$$

Algorithm (device function)

1. Compute d, λ, and the scale $c = \dfrac{\sigma^2(1 - e^{-\kappa\Delta t})}{2\kappa}$.
2. Sample $N \sim \mathcal{P}(\lambda)$ (`curand_poisson`).
3. Sample $G \sim \mathcal{G}(d + N)$ via Marsaglia–Tsang [8] (handles both $\alpha \geq 1$ and $\alpha < 1$).
4. Set $v_{t+\Delta t} = c \cdot G$.

$$\boxed{\text{Poisson: } N \sim \mathcal{P}(\lambda)} \longrightarrow \boxed{\text{Gamma: } G \sim \mathcal{G}(d+N)} \longrightarrow \boxed{v_{t+\Delta t} = c\, G}$$

Scheme 1 — Euler Discretization

Directly discretize both v and S — simplest baseline.

Variance update

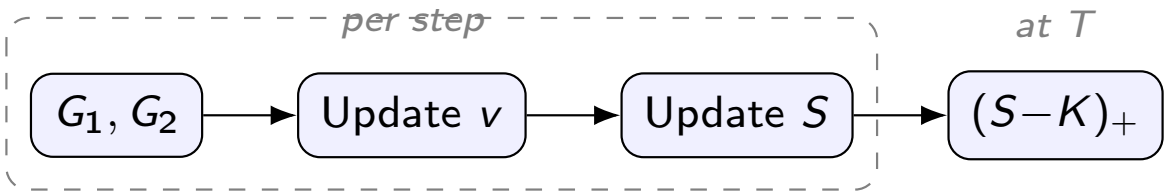
$$\boxed{v_{t+\Delta t} = g\Big(v_t + \kappa(\theta - v_t)\Delta t + \sigma\sqrt{v_t \Delta t}\, G_1\Big)}$$

$g = (\cdot)_+$: truncate to keep $v \geq 0$

Price update

$$\boxed{S_{t+\Delta t} = S_t + rS_t\Delta t + \sqrt{v_t}\, S_t\sqrt{\Delta t}\, \hat{Z}}$$

$$\hat{Z} = \rho\, G_1 + \sqrt{1-\rho^2}\, G_2$$



Scheme 2 — Exact Simulation (Broadie–Kaya)

*Exact CIR variance + closed-form log-price **at maturity**.*

Step 1 — Variance (exact, per step)

$$\boxed{v_{t+\Delta t} = c\,\mathcal{G}(d+N),\ N \sim \mathcal{P}(\lambda)}$$

No truncation needed — exact law

Step 2 (accumulate in loop)

$$\mathtt{vI} \mathrel{+}= \tfrac{1}{2}(v_t + v_{t+\Delta t})\,\Delta t$$

Trapezoidal approx. of $\int_0^T v_s\,ds$

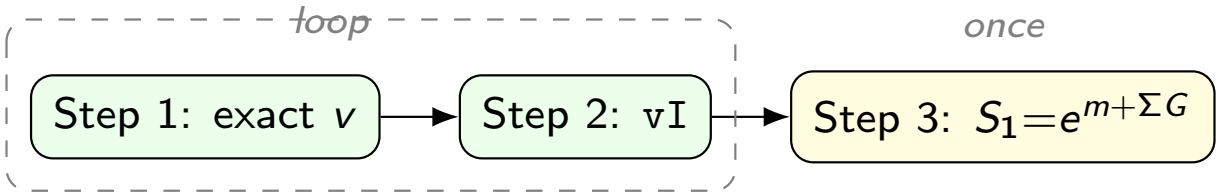
Step 3 — Price (once, after loop)

$$\boxed{S_1 = \exp\big(m + \Sigma\, G\big)}$$

$$\int_0^1 \sqrt{v}\,dW = \tfrac{1}{\sigma}(v_1 - v_0 - \kappa\theta + \kappa\,\mathtt{vI})$$

$$m = -\tfrac{1}{2}\,\mathtt{vI} + \rho\int_0^1 \sqrt{v}\,dW$$

$$\Sigma^2 = (1-\rho^2)\,\mathtt{vI}$$



Scheme 3 — Almost-Exact (Q3)

*Keep exact variance; replace Step-3 with a **per-step** log-price update.*

Variance (same as Scheme 2)

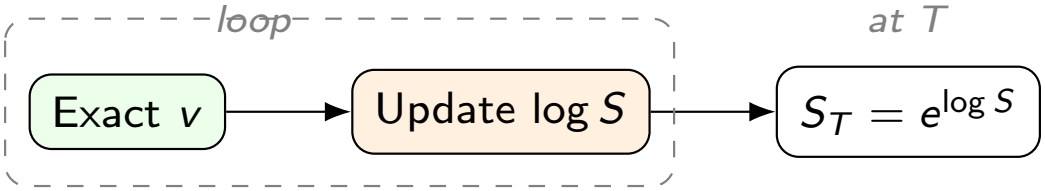
$$\boxed{v_{t+\Delta t} = c\,\mathcal{G}(d+N)}$$

Exact CIR — unchanged

Log-price (per step, approximate)

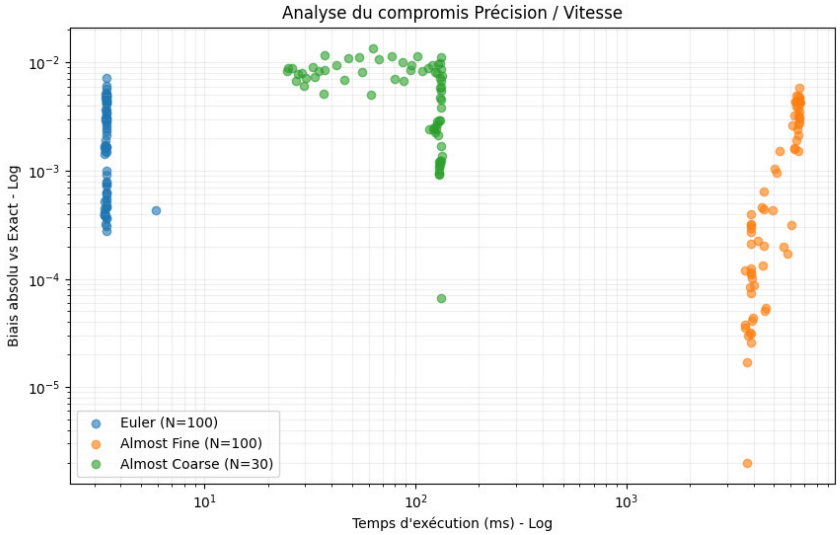
$$\boxed{\begin{aligned}\log S_{t+\Delta t} = \log S_t + k_0 + k_1 v_t + k_2 v_{t+\Delta t} \\ + \sqrt{(1-\rho^2)\,v_t\,\Delta t}\;\hat{G}\end{aligned}}$$

$$k_0 = -\frac{\rho\kappa\theta}{\sigma}\,\Delta t, \qquad k_1 = \left(\frac{\rho\kappa}{\sigma} - 0.5\right)\Delta t - \frac{\rho}{\sigma}, \qquad k_2 = \frac{\rho}{\sigma}.$$

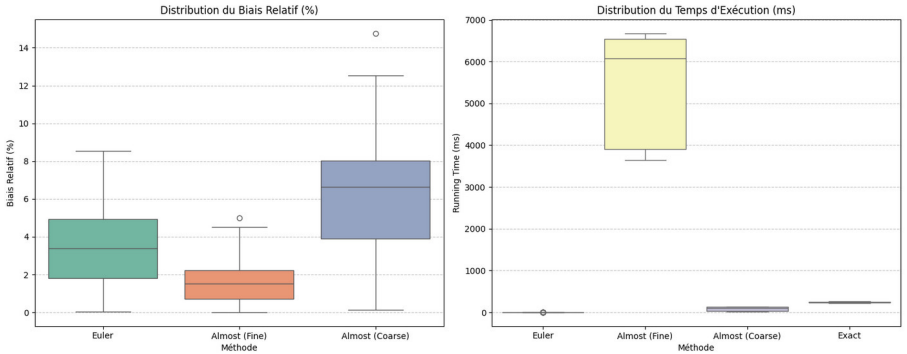


Key advantage: variance exact at *any* Δt $\Rightarrow$ can use Δt=1/30 (30 steps) instead of 1/1000.

Benchmark Results — Visual Analysis



Precision vs Speed (log-log)
Euler (blue) dominates the bottom-left: fast *and* reasonably accurate.



Bias & time distributions
Almost (fine): lowest bias ($\sim$1.5%) but extreme cost ($\sim$6 s per config).

Key insight

Euler is both **faster** and **less biased** than Almost-exact (Δt=1/30). Almost-exact (fine, Δt=1/100) achieves the lowest bias ($\sim$1.5%) but at $\sim$1000$\times$ the cost. The exact CIR transition alone does not compensate for coarse time-stepping in the log-price.